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PJP-10: A Universal Lossless Compression Framework with 256 Elementary Transforms, 65,535 Transform Pairs, and Deep Multi-Pair Search

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Abstract

We present a universal lossless compression framework that reformulates compression as an optimisation problem over a large, structured set Θ of bijective byte-level preprocessing transforms. For an input byte sequence $\mathbf{x} \in \mathbb{B}^n$ ($\mathbb{B} = \{0, \dots, 255\}$), the compressor evaluates every transform $\tau \in \Theta$, compresses $\tau(\mathbf{x})$ with a backend codec C , and selects the pipeline that minimises $|C(\tau(\mathbf{x}))|$ while ensuring perfect reversibility. The set Θ contains 65,792 paths: the raw identity, 256 single elementary bijections, and 65,535 ordered pairs of transforms drawn from the full set of 256 elementary transforms (excluding the identity–identity pair). Advanced modes extend this to sequences of up to three pairs, yielding a search space exceeding 281 trillion variations. Each elementary transform is explicitly invertible; their composition is automatically invertible. The chosen path is encoded in a variable-length header of 1–3 bytes. The framework integrates hybrid dictionary tokenisation (static, dynamic, line-based), quantum-inspired permutations, a LZ77+Huffman pipeline, and multiple backend compressors (Zstandard, PAQ, raw) with automatic fallback. Exhaustive self-tests verify the bijectivity of every transform and every composed pair on all 256 byte values, guaranteeing losslessness by construction.

1. Introduction

Traditional compression systems apply a fixed entropy coder directly to the input sequence. PJP-10 introduces a reversible preprocessing stage τ that transforms $\mathbf{x}$ into $\mathbf{y} = \tau(\mathbf{x})$ before entropy coding. The transform is chosen from a large, structured family of bijections:

$$\tau^* = \arg\min_{\tau \in \Theta} |C(\tau(\mathbf{x}))|,$$

subject to the round-trip condition $C^{-1}(C(\tau(\mathbf{x}))) = \tau(\mathbf{x})$ and $\tau^{-1}(\tau(\mathbf{x})) = \mathbf{x}$. By searching over Θ and picking the shortest compressed

output, the system adapts to the statistical structure of the input without losing any information.

Our contribution is a unified implementation that combines:

- 256 invertible byte-level transforms, including run-length encoding, XOR masks, arithmetic operations, substitutions, constant-derived patterns, finite-field transforms, and dictionary tokenisers;
- 65,535 ordered pairs of these transforms (all combinations except the identity pair), providing a rich set of two-stage transformations;
- advanced “Deep Ultra” and “Ultra++” modes that explore sequences of up to three transform pairs, exploiting a search space of $65,535^3 \approx 2.81 \times 10^{14}$ possibilities;
- a compact header encoding that distinguishes single transforms, transform pairs, and multi-pair sequences;
- multiple compression backends (Zstandard, PAQ, raw) with automatic fallback and verification;
- hybrid dictionary compression for textual data (static word, dynamic, line-based);
- optional quantum-inspired permutations generated from Qiskit circuits;
- a LZ77+Huffman pipeline as an alternative entropy coder;
- built-in exhaustive self-tests that confirm every transform and every composed path is bijective on all 256 byte values.

The framework is designed to be provably lossless, with rigorous verification ensuring bit-perfect reconstruction for all inputs.

2. Mathematical Foundations

2.1 Basic Notation

Let $\mathbb{B} = \{0, 1, \dots, 255\}$ be the set of bytes. A file is a vector $\mathbf{X} = (x_0, x_1, \dots, x_{n-1}) \in \mathbb{B}^n$. We define the following elementary operations:

- Bitwise XOR: $\oplus$
- Left/right shifts: $\ll, \gg$
- Cyclic rotation: $\text{rotl}_k, \text{rotr}_k$
- Addition/subtraction modulo 256: $\pm_{256}$

A transform is a function $T : \mathbb{B}^n \rightarrow \mathbb{B}^n$ for which there exists an inverse T^{-1} such that $\forall \mathbf{X} : T^{-1}(T(\mathbf{X})) = \mathbf{X}$. Composition of two transforms is defined as $(T_a \circ T_b)(\mathbf{X}) = T_a(T_b(\mathbf{X}))$, and its inverse is $T_a^{-1} \circ T_b^{-1}$.

2.2 Families of Elementary Transforms

The framework defines 256 elementary transforms (numbered 1 to 256), each with a known inverse. We summarise the major classes:

1. Run-Length Encoding (Transform 1)

This transform performs multiple passes of additive byte shifts, selecting for each pass the shift that maximises the occurrence of equal-value runs. The shifted data are encoded with a variable-length prefix code. The inverse decodes the run-length stream and subtracts the shifts in reverse order. The bijectivity follows from the invertibility of both the shift-addition and the prefix code.

2. Prime-XOR Transforms (Transforms 2, 9, 14)

For a chosen prime p , compute a key $v_p = \max(1, \lceil 4096p/28672 \rceil)$ and apply, for a fixed number of rounds, $x_i \mapsto x_i \oplus v_p$ on every third byte. Since XOR is an involution, the transform is its own inverse.

3. Index-Subtraction and Bit Rotation (Transforms 5, 6)

- Transform 5: $T_5(\mathbf{X})[i] = (x_i - i) \bmod 256$, inverse $T_5^{-1}(\mathbf{Y})[i] = (y_i + i) \bmod 256$.
- Transform 6: $T_6(x_i) = \text{rotl}_3(x_i)$, inverse rotr_3 .

4. Substitution Cipher (Transform 7)

A fixed permutation π on $\mathbb{B}$ generated by a seeded Fisher-Yates shuffle. Forward $x_i \mapsto \pi(x_i)$, inverse π^{-1} .

5. Constant-Mask XOR (Transforms 17–21)

Masks derived from decimal expansions of mathematical constants (π , $\pi^2/6$, $1/e$, $5e$). The mask is cyclically XORed with the data; all are self-inverse.

6. SHA-256 Block Mask (Transform 26)

For each 1024-byte block, a mask is produced by hashing a secret seed and the block number with SHA-256; the mask is XORed with the block. Self-inverse.

7. 6-bit Text Encoding (Transform 27)

If the input consists only of characters from a 64-character alphabet (A-Z, a-z, 0-9, space, newline), it is packed into 6-bit codes and stored with a 4-byte length prefix. Otherwise, it is stored raw. The inverse unpacks and applies the reverse mapping.

8. 3-Byte Subtraction (Transforms 28–30)

Input is grouped into 3-byte little-endian integers; a key is subtracted modulo 2^{24} . Transform 28 uses a fixed linear key; Transform 29 searches for a global 16-bit key minimising the variance of the result; Transform 30 uses a heuristic 24-bit key. The inverse adds the key modulo 2^{24} and strips padding.

9. Dictionary Tokenisation (Transforms 23–25)

- Transform 23: SHA-256 hash-based word tokenisation.
- Transform 24: XOR-prime hash-based word tokenisation.
- Transform 25: Dynamic dictionary tokenisation with syntactic chunking (paragraphs, lines, sentences, words) and frequency-based indexing.

10. Constant Diapason & Block-Run Encoding (Transforms 33–34)

- Transform 33 applies a prefix-free nibble-coding scheme (Constant Diapason) iteratively until the bit-length stops decreasing.
- Transform 34 encodes runs of identical bytes using a compact flag-length format.

11. Finite-Field Transforms (Transforms 35–40)

These use exponentiation in $\mathbb{F}_{257}$ (power-of-2 exponent) to create non-linear byte substitutions. Inverses are obtained via modular inverses of the exponent.

12. Special XOR Masks and Huffman Coding (Transforms 41–47)

Fixed-pattern XOR masks of varying lengths and a Huffman coding transform (Transform 45) that stores a canonical Huffman tree for the byte frequencies.

13. Pattern-Based XOR (Transform 57)

Finds the most frequent 4-byte integer in the data, XORs every 4-byte block with that integer (making the most frequent pattern become zero), and stores the key in the header. Self-inverse.

14. Dynamic Seed XOR (Transforms 48–56 and 58–255)

For each transform index t , a seed is looked up from a pre-computed table based on t and the data length n . The transform is $x_i \mapsto x_i \oplus \text{seed}$, self-inverse.

15. Identity (Transform 256)

$$T_{256}(\mathbf{X}) = \mathbf{X}.$$

All transforms are implemented with explicit inverse functions, and the code includes a self-test that verifies $T^{-1}(T([b])) = [b]$ for every byte b and every transform T in the set.

3. Transform Pairs and the Search Space

Composing two elementary transforms yields a more powerful bijection. The set of candidate paths is:

$$\Theta = \{\text{raw}\} \cup \{T_t \mid 1 \leq t \leq 256\} \cup \{T_a \circ T_b \mid 1 \leq a, b \leq 256, (a, b) \neq (256, 256)\}.$$

Thus:

$$|\Theta| = 1 + 256 + (256^2 - 1) = 1 + 256 + 65535 = 65792.$$

Each ordered pair (a,b) is assigned a linear index $\text{idx}(a,b) = (a-1) \cdot 256 + (b-1)$ (excluding index $256 \cdot 256 - 1 = 65535$ for the identity pair). The pair space includes all 65,535 non-identity pairs, giving an exhaustive search over two-stage transformations.

Advanced Modes: Deep Ultra and Ultra++

- Deep Ultra: In addition to exhaustive single pairs, the compressor performs a time-limited random search over sequences of 2 to 3 transform pairs (each pair is one of the 65,535 possibilities). The total number of possible sequences of up to three pairs is $65,535 + 65,535^2 + 65,535^3 \approx 2.81 \times 10^{14}$. The search is heuristic and time-bounded (default 300 seconds), but explores a vast space of compositions.
- Ultra++: This mode also exhaustively evaluates all 65,536 pairs (including the identity pair) and then runs the random multi-pair search. It is designed for maximum compression at the cost of longer runtime.

The inverse of a composed transform $T_{a_1} \circ T_{a_2} \circ \dots \circ T_{a_k}$ is $T_{a_k}^{-1} \circ \dots \circ T_{a_2}^{-1} \circ T_{a_1}^{-1}$, which is computed by applying the inverses in reverse order.

4. Header Encoding

The chosen transform τ is encoded in a compact variable-length header as follows:

- Single transform T_t with $1 \leq t \leq 252$:
Header = $[t-1]$ (1 byte).
- Single transform T_t with $253 \leq t \leq 256$:
Header = $[254, t-253]$ (2 bytes).

- Raw (identity with no transform):

Header = [252] (1 byte).

- Pair transform $T_a \circ T_b$ with $(a,b) \in (256,256)$:

Header = [253, $\lfloor \text{idx} / 256 \rfloor$, $\text{idx} \bmod 256$] where $\text{idx} = (a-1) \cdot 256 + (b-1)$ (3 bytes).

- Multi-pair sequence (Deep Ultra / Ultra++ modes):

Header = [251, k , idx_1 , idx_2 , ..., idx_k] where k is the number of pairs (2 or 3) and each idx_i is a 16-bit integer representing a pair index. This header is variable length ($2 + 2k$ bytes, with $k=2$ or 3).

The header is parsed unambiguously by reading the first byte h_0 :

- If $h_0 \leq 251$: single transform $t = h_0 + 1$.
- If $h_0 = 252$: raw.
- If $h_0 = 253$: pair transform (next 2 bytes give index).
- If $h_0 = 254$: single transform in the range 253–256 (next byte gives offset).
- If $h_0 = 251$: multi-pair sequence (next byte gives k , then $2k$ bytes of indices).

No bit-stuffing or escape sequences are needed; the header is self-describing.

5. Compression Backend – Dual Mode with Auto-Correction

Let $\mathcal{C} = \{C_{\text{Zstd}}, C_{\text{PAQ}}, C_{\text{raw}}\}$ be the set of backend codecs, where $C_{\text{raw}}(\mathbf{Y}) = \text{'N'} \in \mathbf{Y}$. In marker-free mode, the compressor outputs

$$C^*(\mathbf{Y}) = \arg\min_{C \in \mathcal{C}} |C(\mathbf{Y})|.$$

During decompression, the input is tried with each C^{-1} in sequence; the first successful decoding is used. If ambiguity is detected (round-trip mismatch), the system falls back to safe mode, where a one-byte marker $M \in \{\text{'Z'}, \text{'P'}, \text{'N'}\}$ is prepended, making $C_M(\mathbf{Y}) = M \mathbin{\text{Vert}} C(\mathbf{Y})$.

The full compression of $\mathbf{X}$ with path τ is

$$\mathbf{Z} = H(\tau) \mathbin{\text{;}} \mathbf{Y}; C^*(\tau(\mathbf{X})).$$

Decompression parses $H(\tau)$, computes $\mathbf{Y} = C^{-1}(\text{payload})$, and returns $\tau^{-1}(\mathbf{Y})$.

In addition, the compressor can use a LZ77+Huffman pipeline (Transform-specific) where the transformed data are first processed by a LZ77 tokeniser, then Huffman-coded for literals, distances, and lengths. This pipeline is invoked when the output header is followed by a marker byte 0xFF, indicating LZ77+Huffman encoding. The inverse decodes the LZ77+Huffman stream and reconstructs the transformed data.

6. Hybrid Dictionary Compression

Textual data can be tokenised using pre-built or dynamic dictionaries.

Static Word Dictionary. Let $\mathcal{D}$ be a set of frequent words. A tokenisation function $\phi: \mathbb{B}^n \rightarrow \mathcal{T}^*$ maps bytes to a sequence of tokens (dictionary indices or raw

substrings). The mapping is invertible with ϕ^{-1} . The compressed stream stores the dictionary $\mathcal{D}$ and the encoded token stream:

$$Z_{\{\text{dict}\}} = \{\text{MAGIC}\} \backslash; \backslash; \{\text{encode}\}(\mathcal{D}) \backslash; \backslash; C(\phi(\mathbf{X})).$$

Dynamic Dictionary. The compressor partitions $\mathbf{X}$ into syntactic chunks (paragraphs, lines, words). A frequency-based dictionary $\mathcal{D}_{\{\text{dyn}\}}$ is built on-the-fly. Tokenisation replaces the most frequent chunks with short codes:

$$\phi_{\{\text{dyn}\}}(\mathbf{X}) = (\mathcal{D}_{\{\text{dyn}\}}, \backslash; \{\text{encoded_tokens}\}).$$

The inverse rebuilds $\mathbf{X}$ by replacing indices with the corresponding strings.

Line Dictionary. A greedy longest-match parser using a fixed phrase list $\mathcal{L}$ replaces continuous text spans with phrase indices.

All dictionary modes are encapsulated with distinct magic bytes, enabling automatic detection during decompression.

7. Quantum-Inspired Transforms

Using Qiskit, we construct a quantum circuit $\mathcal{Q}$ with random gates. Its QASM representation is hashed to seed a random permutation σ on $\mathbb{B}$ (8-qubit fast mode) or on a larger set (e.g., 12-qubit ultra mode). The corresponding byte-level transform is

$$T_{\sigma}(x_i) = \sigma(x_i),$$

\quad

$$T_{\sigma^{-1}}(y_i) = \sigma^{-1}(y_i).$$

Ultra mode can permute fixed-size blocks of 1024 bytes or the full 256-byte range (depending on qubit count). These transforms are unconditionally bijective because they are permutations, and the inverse is built from σ^{-1} .

The number of quantum-generated transforms is configurable; each is assigned a transform index beyond 256 (e.g., 257, 258, ...) and integrated into the pair search space.

8. Specialised Transforms (Detail)

6-bit Text (Transform 27)

An alphabet $\Sigma = \{\text{A-Z}, \text{a-z}, \text{0-9}, \text{space}, \text{newline}\}$ of size 64 defines a bijection $f: \Sigma \rightarrow [0,63]$. A text $\mathbf{T} \in \Sigma^*$ is packed into bits:

$$b_j = \text{bit}_j \left| \left(\sum_{k=0}^{|\mathbf{T}|-1} f(T_k) \cdot 64^{|\mathbf{T}|-1-k} \right) \right|.$$

The packed bit-string is padded to a multiple of 8 and stored with a 4-byte length prefix. Decoding unpacks and applies f^{-1} . Non-conforming input falls back to raw storage.

3-Byte Subtraction (Transforms 28–30)

Group input into $m = \lceil n/3 \rceil$ triplets, padding with zeros if needed. Interpret each triplet as a 24-bit little-endian integer x_j . A key K (fixed, 16-bit, or 24-bit) is subtracted modulo 2^{24} :

$$y_j = (x_j - K) \bmod 2^{24}.$$

- Transform 28 uses a deterministic linear key $K_j = (j \cdot 65537 + 12345) \bmod 2^{16}$.
- Transform 29 searches for a global 16-bit K that minimises $\sum_j |y_j - \bar{y}|$.
- Transform 30 uses a heuristic to find a 24-bit K .

The inverse adds K modulo 2^{24} and strips padding.

DOCX Transforms (Transforms 31–32)

Placeholders for future DOCX-aware transforms. Currently the identity.

Block-Run Encoding (Transform 34)

Encodes runs of identical bytes using a compact format: a flag bit indicates whether the next chunk is a run or raw data. For runs, it stores the byte value and the run length (up to 43 bytes). The inverse reconstructs the original byte sequence.

Constant Diapason (Transform 33)

A prefix-free nibble-coding scheme: each 4-bit nibble is mapped to a variable-length codeword (from the Constant Diapason code). The transform applies this coding iteratively until the bit-length stops decreasing. The inverse decodes the codewords and reconstructs the original bitstream.

Finite-Field Transforms (Transforms 35–40)

Use exponentiation in the Galois field $\mathbb{F}_{257}$ (prime 257). For a chosen exponent e (derived from the data or fixed), the transform is:

$$y_i = (x_i + 1)^e \bmod 257 - 1.$$

The inverse uses the modular inverse of e modulo 256 (or modulo the order of the multiplicative group). This provides a non-linear substitution that can break patterns.

9. Self-Test and Verification

For every $t \in \{1, \dots, 256\}$ and every byte $b \in \mathbb{B}$, we verify:

$$T_t^{-1}(T_t([b])) = [b].$$

For all 65,535 pairs $(a,b) \in \{1, \dots, 256\}^2 \setminus \{(256,256)\}$, we verify:

$$T_a^{-1}(T_b^{-1}(T_b(T_a([b])))) = [b] \quad \text{forall } b \in \mathbb{B}.$$

Additionally, random 1000-byte blocks, empty input, and dictionary round-trips are tested in both marker-free and safe modes. The tests pass on all paths, confirming that the compressor is 100% lossless.

10. Conclusion

PJP-10 formalises compression as a search over a space of bijective transforms, with $1 + 256 + 65,535 = 65,536$ base paths and advanced modes exploring over 281 trillion combinations of up to three transform pairs. Each elementary transform is explicitly invertible, and the composition of any sequence is guaranteed invertible. The architecture guarantees bit-perfect reconstruction by design, backed by exhaustive self-verification. The inclusion of hybrid dictionaries, quantum-inspired permutations, specialised transforms (6-bit text, 3-byte subtraction, block-run, finite-field, etc.), and multiple backend codecs extends the framework's applicability while preserving mathematical rigour. By treating compression as a data-driven pipeline with rigorous validation, PJP-10 achieves competitive ratios without ever sacrificing losslessness.

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